

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 03

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Bandaru Jayasri-192424301 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	20%	15%	15%	
Q1						
Q2						

Signature of the student:

B.Jayasri

Total Marks:

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.
2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.
3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.
4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.
5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.
6. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic

reasoning, design a model to predict safe driving actions. Analyze how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

7. A healthcare organization aims to predict patient risk levels for chronic diseases using historical and lifestyle data. Apply probability learning, Bayes' Theorem, and hypothesis space search to design the model. Analyze the impact of prior probabilities, feature dependencies, and sample size. Evaluate the model's interpretability and its usefulness in preventive healthcare strategies.

8. A bank wants to detect fraudulent transactions using machine learning under uncertain and evolving patterns. Using Bayesian Belief Networks and EM algorithm concepts, design a detection system. Analyze hidden variables, incomplete data handling, and convergence issues. Evaluate how probability-based reasoning improves detection accuracy and reduces false positives in real-world scenarios.

9. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyze how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

10. A research institute is building a system to predict climate changes using historical environmental data. Apply probabilistic models such as Bayesian inference and EM algorithm to handle incomplete and noisy datasets. Analyze convergence behavior, uncertainty modeling, and the influence of prior knowledge. Evaluate the system's effectiveness in long-term environmental forecasting.

11. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

12. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

13. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

14. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply **Naïve Bayes classifiers**, **EM algorithm**, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

15. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using **Bayesian inference** and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

6. Autonomous Vehicle – Bayesian Inference, Naïve Bayes and Probabilistic Reasoning

Answer:

An autonomous vehicle must make decisions under uncertainty because weather, traffic, pedestrians, road conditions, and sensor readings can change continuously. A probabilistic model can combine these uncertain observations and select the safest driving action.

1. Model Design

The system can use **Bayesian inference** to estimate the probability of different driving situations.

For example:

- Weather → Rainy / Clear / Foggy
- Traffic → Low / Medium / High
- Pedestrian → Present / Absent
- Road condition → Safe / Slippery

- Action → Brake / Slow down / Maintain speed / Change lane

Bayes' Theorem is:

$$P(H|E)=P(E)P(E|H)P(H)$$

where H is a possible driving situation and E represents sensor observations.

A **Naïve Bayes classifier** can classify situations such as:

- Safe to continue
- Slow down
- Brake
- Change lane

It assumes that features are conditionally independent given the class.

2. Handling Uncertainty

Sensor readings are never perfectly accurate. Cameras, radar, LiDAR and GPS may produce noisy or incomplete information.

Bayesian reasoning represents uncertainty using probabilities rather than fixed decisions. For example:

- Probability of pedestrian = 0.85
- Probability of clear road = 0.20
- Probability of collision = 0.75

The vehicle can therefore select a safer action when the probability of danger becomes high.

3. Handling Sensor Noise

Multiple sensors can be combined to improve reliability. If the camera detects an object but LiDAR gives uncertain information, Bayesian inference can combine both observations and update the probability.

Continuous sensor observations allow the system to update its beliefs dynamically.

4. Reliability in Dynamic Conditions

The model can continuously update predictions when:

- Weather changes
- Traffic density increases
- A pedestrian enters the road
- A vehicle suddenly brakes
- Sensor readings change

A safety threshold can be defined. For example, if the estimated collision probability exceeds a certain threshold, the vehicle can automatically slow down or brake.

5. Advantages

- Handles uncertainty mathematically.
- Works with noisy sensor data.
- Can update predictions using new observations.
- Naïve Bayes is computationally efficient.
- Supports real-time decision-making.

6. Limitations

- Naïve Bayes assumes conditional independence, which may not always be true.
- Incorrect sensor data can affect predictions.
- Complex environments may require more advanced probabilistic models.
- Real-time systems require high computational efficiency and very low prediction latency.

Conclusion

A combination of **Bayesian inference**, **Naïve Bayes** and **probabilistic reasoning** provides a practical approach for autonomous driving. It allows the vehicle to estimate risks, handle sensor uncertainty and dynamically select safer actions. However, extensive training, sensor fusion and real-time validation are required to achieve high reliability.

7. Healthcare Risk Prediction – Probability Learning, Bayes' Theorem and Hypothesis Space Search

Answer:

A healthcare organization can develop a probabilistic machine learning system to predict a patient's risk of developing a chronic disease such as diabetes or heart disease. The system can use historical medical and lifestyle data such as age, blood pressure, BMI, glucose level, smoking, exercise and family history.

1. Model Design

The system can use **Bayes' Theorem** to calculate the probability of disease based on patient characteristics.

$$P(D|X)=P(X)P(X|D)P(D)$$

where:

- D = disease/risk condition

- X = patient's observed features
- $P(D)$ = prior probability of disease
- $P(X|D)$ = probability of observing the features given disease
- $P(D|X)$ = posterior probability of disease

The output can classify patients into:

- Low Risk
- Medium Risk
- High Risk

2. Probability Learning

The system learns probabilities from historical patient data.

For example, if the training data shows that patients with high glucose and high BMI frequently develop diabetes, the model learns a higher probability for that combination.

As more patient data becomes available, the estimated probabilities can be updated.

3. Impact of Prior Probabilities

Prior probability represents existing knowledge before considering the patient's current features.

For example, if a disease is common in a particular population, its prior probability will be higher. If the disease is rare, the prior probability will be lower.

Bayesian inference combines this prior knowledge with current patient information to produce a posterior probability.

4. Feature Dependencies

Medical features are often dependent on each other. For example:

BMI $\rightarrow$ Blood Pressure $\rightarrow$ Heart Disease Risk

Similarly:

Age + Family History + Glucose Level $\rightarrow$ Diabetes Risk

A simple Naïve Bayes model assumes feature independence, which may not accurately represent these relationships. A **Bayesian Belief Network (BBN)** can be used when dependencies between medical variables need to be explicitly represented.

5. Hypothesis Space Search

The hypothesis space contains possible rules or models that can classify patients.

For example:

- H1: High glucose $\rightarrow$ High risk
- H2: High glucose + high BMI $\rightarrow$ High risk
- H3: High glucose + high BMI + family history $\rightarrow$ High risk

The learning algorithm searches for a hypothesis that provides good predictions on training data while also generalizing to unseen patients.

6. Effect of Sample Size

A larger training dataset generally provides more reliable probability estimates.

Small sample:

- Probability estimates may be unreliable.
- Higher risk of overfitting.
- Poor generalization.

Large sample:

- More accurate probability estimates.
- Better representation of different patient groups.
- Improved generalization.

Therefore, sufficient representative training data is important.

7. Interpretability

One major advantage of Bayesian models is interpretability. Healthcare professionals can understand why a patient received a particular risk prediction.

For example:

High glucose + high BMI + family history $\rightarrow$ High posterior probability of diabetes.

This makes the system more useful for clinical decision support.

8. Usefulness in Preventive Healthcare

The model can identify high-risk patients at an early stage. Healthcare professionals can then recommend:

- Regular health checkups
- Diet modifications
- Physical activity
- Monitoring of blood glucose
- Early medical intervention

9. Limitations

- Poor-quality data can produce incorrect predictions.
- Biased training data can affect certain patient groups.
- Feature dependencies may violate the assumptions of simple Naïve Bayes.
- Medical predictions should support, not replace, professional clinical judgment.
- Privacy and security of patient data must be maintained.

Conclusion

A combination of **Bayes' Theorem, probability learning and hypothesis space search** can effectively predict chronic disease risk. Prior probabilities and feature relationships strongly influence predictions, while larger and representative datasets improve reliability. The interpretability of probabilistic models makes them valuable for preventive healthcare and early risk identification.